FVGs + IFVG + OTE Indicator — Design Document

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Overview

"FVGs+IFVG+OTE5" is a Pine Script v6 overlay for TradingView. It draws fair value gaps (FVGs), detects inverse FVGs (IFVGs) when price closes through them, and flags the IFVGs that line up with an ICT Optimal Trade Entry (OTE, the 0.62–0.79 retracement of a leg) on the chart timeframe or on the next two higher timeframes.

Term	Meaning in this script
FVG	Three-candle gap, built from the candles at [1], [2], [3]. "Long" (bullish) when the low of [1] is above the high of [3]; "short" (bearish) when the high of [1] is below the low of [3]. The drawn zone is widened to candle bodies when a body gap exists (suspension blocks).
1PFVG	First presented FVG after the session open (09:30 New York, 09:00 Madrid for DAX), tracked only on minute charts up to 3m. Drawn in its own colors and extended to the right.
1PFVG inverse	The first FVG after the 1PFVG with the opposite bias (optional).
IFVG	An FVG that price closes through. A long FVG closed below its low becomes a bearish IFVG ("IFVG-"); a short FVG closed above its high becomes a bullish IFVG ("IFVG+").
OTE	Retracement band between oteMin and oteMax (0.62 / 0.79 by default) of a leg. An IFVG is an "OTE IFVG" when price rejected the OTE of the leg before the FVG, or when it retraces into an IFVG that sits in the OTE of the leg since the inversion.
C.E.	Consequent encroachment: the midpoint of an FVG or of a candle wick.

What it produces:

- FVG boxes (or no-fill boxes, or high / low / C.E. segments), with mitigated FVGs closed at the bar that mitigated them.
- IFVG candidates and confirmed IFVGs, drawn on the FVG's own box and labelled "IFVG+", "IFVG-" or "IFVG± OTE <timeframes>".
- OTE bands drawn to the right of the last candle, one per tracked timeframe.
- Wick C.E. segments, optional signal markers and alerts.

Execution model

The script runs once per bar, top to bottom, and keeps all state in `var` globals and in arrays of user-defined types. Everything that confirms something (IFVG confirmation, signals) is evaluated on closed bars only (`barstate.isconfirmed`).

Per-bar pipeline, in source order:

1. **Inputs and constants.** Draw-mode flags (`fponly`, `ifvgOnly`, `hideReg`), filter flags (`fltOff`, `fltMid`) and the fixed filter values `flN = 1` and `flWin = 25`.
2. **Wick C.E.** If enabled, the wick of candle [1] is measured and a C.E. segment is drawn; the oldest is deleted beyond `nwicks`.
3. **Session setup.** `tz = Auto` resolves to Madrid (DAX) or New York; the 1PFVG time is parsed into `fph` / `fpm`.
4. **FVG detection.** A long and a short FVG are computed from candles [1]–[3]. Each new FVG becomes an `FVG` object and is classified: 1PFVG, 1PFVG inverse, or regular. Regular FVGs go through `route()` (merge and filter); in "1PFVGs" draw mode they are discarded.
5. **Sweep detection.** Short-term pivots (`flL`, `flR`) are tracked; a bar breaking the last one calls `onSweep()`, which reveals pending FVGs.
6. **FVG update.** `updfvgs()` creates or updates boxes, lines and C.E. lines, closes mitigated FVGs and fires bounce alerts, for the long, short and first-presented lists.
7. **Pivots for OTE.** Chart-timeframe pivots (`pivL`, `pivR`) and, through `request.security`, pivots of the next two timeframes on the ladder.
8. **IFVG candidate selection.** Close mode: every new long / short FVG replaces the bearish / bullish candidate. Swing mode: a significant swing pivot picks the last FVG before it. The candidate's FVG box is adopted as the IFVG box (`adopt`).
9. **Confirmation and cancellation.** A close through the FVG confirms the IFVG, labels it and starts the leg measurement; a close back through it cancels. Close mode signals on the confirming bar; swing mode signals on a later retest.
10. **Kept IFVGs.** `updKept()` extends kept boxes and restores invalidated ones.

11. **OTE step.** `gStep()` runs the three trackers of each of the three groups (`gBear` , `gBull` , `gSw`) and returns an OTE signal direction.
12. **Output.** OTE labels and highlight, `plotshape` markers, then the alerts.

Inputs

The settings dialog has six sections, ordered so that what to draw and which FVGs to keep come first. Inputs that share an `inline` id sit on one line. Defaults are in brackets.

Section	Inputs	Effect
Enable / Draw / Filter	`dofp` 1PFVGs [on], `doifvg` IFVGs [on], `dowick` Wick C.E. [on]; `drawMode` [all]; `fltMode` [none]	`drawMode` is all, 1PFVGs, IFVGs or 1PFVGs+IFVGs. The IFVG modes hide regular FVGs, wicks, swing labels and signals; alerts keep working. `fltMode` is none, sweeps or sweeps&mid.
FVG Settings	`visible` Historic compute [off], `dofvgalert` ; `minfvg` [3], `maxfvg` [10000], `maxfvgs` keep [200]; `dupTol` merge tolerance [10%], `flL` / `flR` sweep pivots [3 / 3]; `drawStyle` [Box]; C.E. height / width / style; FVG colors	Size limits are in points. `dupTol = 0` turns merging off. `drawStyle` is Box, Box no fill, or Segments (hi / lo / C.E.).
1st Presented FVGs	`fpfvgtime` [0930], `tz` [Auto], `nfp` keep [6]; day / week / inverse colors; `doifp` [off]	Session open for the 1PFVG; the week variant (Monday) has its own color.
Wick Settings	`nwicks` [30], `minwick` [20], `wickdx` [0 = auto], style, color, `dowickalert`	Wick C.E. segments and their bounce alerts.
IFVG Settings	Alerts, signals, labels; `keepIFVG` + `keepMax` [off / 20]; `ifvgswings` [off], significant swing, swing labels; `pivL` / `pivR` [10 / 3]; `atrLen` [14], `minATR` [4]; IFVG colors, transparency, text size and alignment	`ifvgswings` switches from the close-mode logic to the swing logic. `pivL` / `pivR` also set the pivots that find the leg before the FVG for OTE.

Section	Inputs	Effect
IFVG + OTE	`useOTE`, `oteOnly`, `showOTEsig`; `oteMin` / `oteMax` [0.62 / 0.79], `oteMode` [Overlap]; prior-leg rejection (`oteRej`, `oteRejAny`, `oteRejCap`); `oteHtf1`, `oteHtf2`, `oteHtfL` / `oteHtfR` [5 / 2]; band draw, offset [5], width [8]; band, border and text colors	`oteMode` is Overlap, Fully inside or C.E. inside. The band is drawn `oteOffset` candles right of the last candle and is `oteWidth` candles wide.

Two values are not inputs but globals next to the filter inputs: `flN = 1` (FVGs shown after a sweep) and `flWin = 25` (the mid-arm window, as % of the arm length on each side of its middle).

Data structures

Five user-defined types carry the state. Pine cannot reassign globals inside functions, so every function mutates object fields instead.

FVG

One object per detected FVG, whether or not it is drawn.

Field	Role
`hi`, `lo`, `ce`	Zone and midpoint. Widened in place by merges.
`islong`	Bullish (long) FVG.
`start`, `end`	Start time (time of candle [3]); `end` is the mitigation time, or `na` while open.
`sbar`	Bar index where the FVG starts, used for the OTE prior leg and the mid-arm filter.
`rect`, `celine`, `hiln`, `loln`	Drawing handles: box, C.E. line, high and low segments.
`isfp`, `iswfp`, `isifp`	1PFVG, week 1PFVG, inverse 1PFVG.
`dup`	Older almost identical FVG that absorbed this one.
`kept`, `kdir`	The box is currently shown as an IFVG, and its direction (-1 bearish, 1 bullish).

Field	Role
`cand`	Current IFVG candidate, not yet released.
`own`	The box exists only because of the IFVG (the FVG itself is not drawn).
`shown`	The FVG is in the drawn lists.

FltState

State of the sweep filter, held in the single object `fs`: `post` (FVGs shown since the last sweep, starts at 1,000,000 so nothing is shown before the first sweep), `stH` / `stL` (last unswept short-term high / low) and their bars `stHbar` / `stLbar`.

PivSet

Pivots of one timeframe: parallel arrays `hb` / `hp` (pivot-high chart bar and price) and `lb` / `lp` (pivot lows), a `tag` such as "5m", and `htf`. At most 60 pivots are kept. Three instances exist: `pv0` (chart), `pv1` and `pv2` (next two higher timeframes). Higher-timeframe pivots are stored with the chart bar they map to.

OteTrack

Follows one IFVG against the pivots of one timeframe.

Field	Role
`dir`, `zlo`, `zhi`	Direction and IFVG zone.
`legHi`, `legLo`, `rext`	The leg since the inversion, and the running retrace extreme since the FVG.
`armed`, `fired`	IFVG confirmed; OTE signal already fired for this confirmation.
`ctime`, `cbar`	Confirmation time and bar.
`pHi`, `pLo`, `rejHit`, `rej`	Prior leg, whether the confirming candle rejected its OTE, and the band drawn for it.
`band`, `tag`, `sbar`	Retrace OTE band, timeframe label, FVG start bar.

OteGroup

Three trackers (`t0` chart, `t1` , `t2`) on the same IFVG, plus `flagged` (once per IFVG) and `tags` (timeframes that matched, e.g. "5m+15m"). Three groups exist: `gBear` and `gBull` for close mode, `gSw` for swing mode.

Global state

Variable	Content
`lfvgs` , `sfvgs`	Drawn long and short FVGs, sorted by start.
`fpfvgs`	1PFVGs and inverse 1PFVGs (at most `nfp`).
`pendF`	FVGs created since the last sweep and not shown (at most 60).
`keptIf`	FVGs whose box is currently shown as an IFVG.
`bearifvg` , `bullifvg` , `bearconfirm` , `bullconfirm`	Close-mode candidates and their confirmation times.
`ifvgBox` , `ifvgBoxb`	Boxes of the bullish and bearish IFVG under watch.
`swFvg` , `ifstime` , `ifshi` , `ifslo` , `ifsdir` , `ifsconfirm`	Swing-mode IFVG under watch.
`wicks`	Wick C.E. lines.

Functions and methods

Pine requires every function to return a value, so functions that only mutate state end with `true` .

FVG detection and drawing

Function	Role
`isLongFVG` , `isShortFVG`	Gap tests: `l1 > h3` and `h1 < l3` .
`longFVGHL` , `shortFVGHL`	Return the drawn zone extended to bodies, plus the wick-based zone used for the size filter.

Function	Role
`isFPFVG`	True inside the 1PFVG window: hour = `fph`, minute beyond `fpm` plus two candles, minute chart up to 3m.
`clearlastfp`	Keeps at most `nfp` 1PFVGs; the oldest stops extending and is killed.
`updfvgs(array)`	Per FVG: closes it when mitigated, creates its box or segments and C.E. line, follows the right edge, fires bounce alerts, and trims the array beyond `maxfvgs`.
`bgColor`, `lnColor`, `endsAt`, `isDone`, `kill`, `alerts`	Methods on `FVG`: colors (with the hidden-by-draw-mode rule), end time, mitigation test, delete drawings, bounce alerts.
`bounces(p)`	True when the bar touches level `p` and closes away from it.

Filtering and merging

Function	Role
`route(FltState, FVG)`	Entry point for a regular FVG: merge into a duplicate, or show it, or put it in `pendF`.
`findDup(FVG)`	Most recent drawn FVG (any bias) within the merge tolerance, searching `lfvgs`, `sfvgs`, `fpfvgs` and `keptIf`.
`mergeZone(g, f)`	Widens both FVGs to max high / min low, moves the C.E., updates box and lines, and re-opens `g`. Frozen when `g` is a kept IFVG.
`insertSorted(array, FVG)`	Inserts by start time and marks the FVG as shown.
`reveal(FVG)`	Shows a pending FVG, or merges it into an existing duplicate (the merged box then starts at the earlier start).
`onSweep(armStart)`	Reveals the last pending FVG and, with `sweeps&mid`, the largest one near the middle of the arm; clears `pendF`.

IFVG box handling

Function	Role
`adopt(FVG, dir)`	Turns the FVG's box (or its duplicate's) into the IFVG box; creates one if the FVG has none. Returns the box.
`release(FVG, confirmed, ctime)`	Candidate replaced: keep the box as IFVG if confirmed and `keepIFVG`, otherwise restore it.
`restore(FVG)`	Back to a plain FVG: deletes an own box, or restores colors, width, text and extension.
`updKept()`	Extends kept boxes to now; restores those the price closed back through; drops dead entries.
`findFVG(time, array)`	Swing mode: last FVG starting before a swing time.

OTE

Function	Role
`tfName`, `tfstr`	Timeframe labels for tags and alerts.
`addHigh`, `addLow`, `lastHigh`, `lastLow`	Methods on `PivSet`.
`barOfTime(t)`	Chart bar index of the candle at time `t`, scanning back at most 4,900 bars.
`feedHtf`	Registers a new higher-timeframe pivot once, on the first chart bar after its bar closed.
`priorLeg(dir, sbar, set)`	Returns `[legHigh, legLow]` of the leg before the FVG.
`oteStart`, `oteConfirm`, `oteCancel`, `oteStep`	Tracker life cycle: start watching a candidate, confirm it, cancel it, evaluate one bar.
`gStart`, `gConfirm`, `gCancel`, `gStep`	The same actions applied to the three trackers of a group; `gStep` returns the signal direction and builds the tags.

Helpers

`getstyle`, `getsize`, `gethalign`, `getvalign` map input strings to Pine constants. `wdx` returns the wick segment width. `hmParse` splits "0930" into hour and minute. `timeat` reads a future-offset time.

FVG detection, filtering and life cycle

Detection

On every bar the script looks at candles [1], [2] and [3]. A gap is valid when its wick-based height lies between `minfvg` and `maxfvg`. The stored zone is the wick gap, widened to a candle body when a body gap exists next to it. The FVG starts at the time of candle [3] and `sbar = bar_index - 3`. The newest long and short FVGs of the bar are kept in `lastlongfvg` and `lastshortfvg`, because the close-mode IFVG logic uses them even when they are later hidden or merged.

Classification of a new FVG, in order:

1. If a 1PFVG exists and the inverse option is on, the first FVG of the opposite bias is the inverse 1PFVG.
2. Otherwise, if no 1PFVG exists yet and the bar is in the 1PFVG window, it is the 1PFVG (`iswfp` on Mondays).
3. Otherwise it is a regular FVG: `route()` handles it, or it is discarded when `drawMode` is 1PFVGs.

Life cycle

An FVG is mitigated when a candle closes or opens beyond its far side (below `lo` for long, above `hi` for short). Its `end` is then fixed and the box stops growing. Each array is trimmed to `maxfvgs` by deleting the oldest. 1PFVGs and inverse 1PFVGs extend to the right until `clearlastfp` retires them.

Sweep filter

With `fltMode = none`, every regular FVG is drawn. With `sweeps` or `sweeps&mid`, a regular FVG is drawn only when it matters around a liquidity sweep:

- Short-term highs and lows are pivots with `flL` / `flR` bars. A sweep is a bar whose high is above the last unswept short-term high (or whose low is below the last short-term low), tested against levels known before that bar.

- FVGs that are not shown wait in `pendF`. On a sweep, `onSweep()` shows the last one (the FVG created just before the sweep) and resets `fs.post` to 0, so the next `flN = 1` FVG is shown at creation.
- With `sweeps&mid`, `onSweep()` also shows the largest pending FVG whose start lies within `flWin = 25` % of the arm length around the arm's middle. The arm runs from the last opposite short-term pivot to the sweep bar.
- Before the first sweep nothing is shown. 1PFVGs and the close-mode IFVG logic are not filtered.

Merging near-identical FVGs

A new FVG, of any bias, whose high and low are each within `dupTol` % of its height of an existing FVG is not drawn. The existing one absorbs it:

1. `findDup()` returns the most recent match among the long, short, first-presented and kept-IFVG FVGs.
2. `route()` sets `f.dup` and calls `mergeZone()`: both FVGs take the max high and min low, and the old box, lines and C.E. are moved. If the old FVG was mitigated it is re-opened.
3. The new FVG is not stored in the drawn lists, so it is never drawn, but it still feeds the IFVG logic: `adopt()` and `release()` redirect to `f.dup`.
4. The same check runs again in `reveal()` when a pending FVG is shown, because the zone may have been drawn in the meantime. If the revealed FVG is the older one, the merged box starts at its start time.
5. A kept IFVG is never widened: its zone is frozen, and the new FVG is simply not drawn.

IFVG logic

An IFVG goes through three states: candidate (under watch), confirmed (a close went through the FVG) and cancelled (a close came back through it). Two modes choose the candidate.

Close mode (default)

- **Candidate.** Each new long FVG becomes the bearish candidate `bearifvg`; each new short FVG becomes the bullish candidate `bullifvg`. The previous candidate of that side is released.
- **Confirmation.** On a closed bar, `close < bearifvg.lo` confirms a bearish IFVG ("IFVG-"); `close > bullifvg.hi` confirms a bullish one ("IFVG+"). The confirming bar is also the signal bar (`sig = -1` or `1`).

- **Cancellation.** After confirmation, a close back above `bearifvg.hi` (or below `bullifvg.lo`) clears the confirmation and the label, and cancels the OTE trackers. The candidate stays under watch.

Swing mode

One IFVG is under watch at a time.

- **Candidate.** A pivot (`pivL` / `pivR`) is significant when its distance to the last opposite pivot is at least `atr * minATR`, or when the swing filter is off. A significant pivot high selects the last long FVG that starts before it (bearish IFVG); a pivot low selects the last short FVG (bullish IFVG). The swing price is the origin of the OTE leg.
- **Confirmation and cancellation.** Same close rules as close mode, evaluated on every closed bar.
- **Signal.** After confirmation, on a later bar, price returns into the IFVG: for a bearish IFVG `high >= ifslo` and `high < ifshi`; for a bullish one `low <= ifshi` and `low >= ifslo`.
- Candidates are taken from `lfvgs` / `sfvgs`, so in this mode the sweep filter and the merge also decide which FVGs can become IFVGs.

One box per IFVG

The IFVG never gets a second box next to its FVG; the FVG's own box is restyled.

- `adopt()` styles the box as an IFVG: IFVG colors, border width 2, no extension, right edge at the current time. If the FVG has no box (hidden by a draw mode or the filter, or absorbed by a duplicate), a box is created on the FVG, or on its duplicate, and flagged `own`.
- If the FVG is also a 1PFVG or an inverse 1PFVG, the box keeps the 1PFVG colors, border width and right extension. Only the text changes.
- `release()` runs when the candidate is replaced. With `keepIFVG` on and the IFVG confirmed, the box stays an IFVG box, its FVG gets `end = confirmation time`, and it joins `keptIf`. Otherwise `restore()` returns the box to a plain FVG, or deletes it if it was `own`.
- `updKept()` runs every bar: it extends kept boxes to now, and restores a kept IFVG when a confirmed close goes back through it (above `hi` for bearish, below `lo` for bullish). When `keptIf` exceeds `keepMax + 3`, the oldest non-candidate is restored.

Labels are set where the state changes: "IFVG-" / "IFVG+" on confirmation, empty on cancellation, "IFVG± OTE <timeframes>" with the `oteHighlight` border on an OTE signal. The label color is transparent when `Labels` is off.

OTE logic

Each IFVG under watch is followed by three `OteTrack` objects: the chart timeframe and the next two higher timeframes. An OTE on any enabled timeframe flags the IFVG, once per IFVG, and the matching timeframes are listed in the label (for example "5m+15m"). Each tracker runs two independent checks.

Timeframe ladder

The ladder is 3m (or lower), 5m, 15m, 1h, 4h, D. The chart sits on the highest rung at or below its timeframe, and the next two rungs are HTF1 and HTF2 (absent at the top of the ladder). Higher-timeframe pivots come from `request.security` with `lookahead_on` and `[1]` offsets, so they only use closed higher-timeframe bars and do not repaint. A new pivot is registered once, when the pivot time changes, and `barOfTime()` maps it to a chart bar.

Check 1: rejection of the prior leg

When the IFVG is confirmed, the leg before the FVG is measured with `priorLeg()`:

- Bearish IFVG: the last pivot high before the FVG start, down to the lowest low up to the FVG start (a prior down leg).
- Bullish IFVG: the last pivot low before the FVG start, up to the highest high up to the FVG start (a prior up leg).

The OTE band of that leg is measured from the leg's end: for a bearish IFVG, `pLo + oteMin * range` to `pLo + oteMax * range`. The confirming candle is a rejection when its wick (its high for bearish, its low for bullish) reaches into that band and the close ends back away from it. `oteRejAny` also counts the running extreme of any candle between the FVG and the confirmation. `oteRejCap` requires the wick not to pass the upper OTE level; otherwise it may go deeper unless it takes out the leg origin. On a hit, a dashed band is drawn and the signal fires on the confirming bar. The IFVG itself need not be inside the band.

Check 2: retrace into the IFVG

After confirmation, the leg is measured from its origin to the extreme reached since the inversion:

	Bearish IFVG	Bullish IFVG
Origin	Highest high since the FVG (close mode: `h4` and later bars; swing mode: the swing high)	Lowest low since the FVG (close mode: `l4`; swing mode: the swing low)
Extreme	Lowest low since confirmation	Highest high since confirmation
OTE band	`legLo + oteMin * r` to `legLo + oteMax * r`	`legHi - oteMax * r` to `legHi - oteMin * r`
Tap	`high >= zlo`	`low <= zhi`

On higher timeframes the origin is moved to the latest higher-timeframe pivot when that pivot is beyond the retrace extreme. The IFVG matches the band according to `oteMode`: any overlap, fully inside, or its midpoint inside. The signal fires when the IFVG matches and price taps back into it on a later bar than the confirmation, and not if the signal already fired for this IFVG. The extreme is updated after the bar is evaluated, so the bar being tested never defines its own leg.

Bands and signals

Bands are drawn right of the last candle, `oteOffset` candles away and `oteWidth` candles wide, and are moved every bar. A band is yellow while the IFVG matches the OTE and gray while it does not. When a group raises a signal, `oteSig` is -1 or 1, a diamond is plotted if `showOTEsig` is on, the IFVG label becomes "IFVG± OTE <timeframes>", and the OTE alert fires.

Drawing, alerts, limits and known risks

Drawing

- **Styles.** Box; Box, no fill (border only); Segments (high, low and C.E. lines, no box).
- **Draw modes.** In the IFVG modes, regular FVGs are drawn fully transparent rather than deleted, so they keep feeding the logic. A 1PFVG or IFVG box stays visible where the mode asks for it.
- **Limits.** The script allows 500 boxes and 500 lines. Segments use two or three lines per FVG, so fewer FVGs stay visible in that style. A box that is drawn only for an IFVG counts against the same limit.

Alerts

Alert	Condition
Bounce on FVG hi, lo or C.E.	`dofvgalert`, price touches the level of an open FVG and closes away; once per bar.
Bounce on wick C.E.	`dowickalert`.
"LONG IFVG <tf>" / "SHORT IFVG <tf>"	`doifvgalert` and an IFVG signal; suppressed when `oteOnly` is on.
"LONG IFVG+OTE <tf>" / "SHORT IFVG+OTE <tf>"	`doifvgalert`, `useOTE` and an OTE signal.

Design decisions

- **One box per IFVG.** Adopting the FVG box avoids two boxes in the same place, at the cost of the extra state fields (`kept`, `cand`, `own`, `dup`) and the restore path.
- **Merged FVGs stay objects.** An absorbed FVG is not drawn but still feeds the IFVG logic, so IFVG detection does not depend on what is visible.
- **Fixed filter values.** One FVG before and after a sweep and a 25% mid window are constants, not inputs.
- **Swing-mode confirmation** is evaluated on every bar. The original code evaluated it only on pivot-low bars, which is why it was moved to the top level.

Known risks and limits

- Merging two FVGs when the target already hosts another IFVG makes them share one box.
- A kept IFVG box is deleted if its FVG is shifted out of the `maxfvgs` window.
- `findFVG()` returns the oldest FVG in the list when none starts before the swing time.
- `barOfTime()` scans back up to 4,900 bars and `priorLeg()` up to 4,000 bars, on each new higher-timeframe pivot or IFVG start; the script declares `max_bars_back = 5000`.
- Higher-timeframe OTE needs the two requested timeframes to be higher than the chart, so near the top of the ladder one or both trackers are absent.
- Historic compute is off by default, so FVGs older than the last `maxfvgs` are not shown when scrolling back.